

PID Controller Design Project

AE 432: Flight Dynamics and Control

Instructor: Dr. K. Merve Dogan

Authors: Anuranan Bharadwaj, Jose Garcia, Khoa Nguyen,
Justin Noh, Kalkamanali Satvaldy, Vincent Shi

Embry-Riddle Aeronautical University

Daytona Beach, FL, 32114-3800

December 4th , 2025



I. Aircraft Stability Characteristics

The McDonnell Douglas DC-8 was selected as the aircraft for the controller design project. The DC-8 is a jet airliner introduced during the 1950s and helped to establish long-range subsonic travel. Attached below are tables of the aircraft specifications and stability characteristics for the

McDonnell Douglas DC-8:

McDonnell Douglass DC-8

Aircraft Parameter Assume $\alpha_{L=0} = -2^\circ$	DC-8	DC-8	DC-8
Altitude (ft)	S/L	15,000	33,000
Mach	0.218	0.443	0.88
True Airspeed (ft/s)	243	468	868
Dyn Pressure (lb/ft ²)	70.4	164	300
Weight (lb)	190,000	190,000	230,000
Wing Area— S —(ft ²)	2,600.0	2,600.0	2,600.0
Wing Span— b —(ft)	142.3	142.3	142.3
Wing Chord— $\bar{c}$ —(ft)	23.0	23.0	23.0
C.G. (x $\bar{c}$)	0.15	0.15	0.15
Trim AOA (deg)	10.36	3.23	0.45
I_{xx_S} (slug — ft ²)	3.16×10^6	3.13×10^6	3.77×10^6
I_{yy_S} (slug — ft ²)	2.94×10^6	2.94×10^6	3.56×10^6
I_{zz_S} (slug — ft ²)	5.51×10^6	5.86×10^6	7.13×10^6
I_{xz_S} (slug — ft ²)	-4.14×10^5	-2.20×10^5	2.73×10^5
Longitudinal Derivatives			
X_u (1/s)	-0.0291	-0.0071	-0.0463
X_α (ft/s ²)	15.32	15.03	-22.36
Z_u (1/s)	-0.251	-0.133	0.062
Z_α (ft/s ²)	-152.8	-354	-746.9
M_u (1/ft · s)	0.0000	0.0000	-0.0025
M_α (1/s ²)	-2.118	-5.010	-12.003
M_β (1/s)	-0.260	-0.337	-0.449
M_q (1/s)	-0.792	-0.991	-1.008
X_{δ_e} (ft/s ²)	0.0	0.0	0.0
Z_{δ_e} (ft/s ²)	-10.17	-23.70	-39.06
M_{δ_e} (1/s ²)	-1.351	-3.241	-5.120
Lat-Dir Derivatives			
Y_β (ft/s ²)	-27.05	-47.19	-81.28
L_β (1/s ²)	-1.334	-2.684	-5.111
L_p (1/s)	-0.949	-1.234	-1.299
L_r (1/s)	0.611	0.391	0.352
N_β (1/s ²)	0.762	1.272	2.497
N_p (1/s)	-0.119	-0.048	-0.008
N_r (1/s)	-0.268	-0.253	-0.254
Y_{δ_r} (ft/s ²)	5.782	13.47	20.35
L_{δ_r} (1/s ²)	0.185	0.375	0.639
N_{δ_r} (1/s ²)	-0.389	-0.861	-1.298
Y_{δ_a} (ft/s ²)	0.0	0.0	0.0
L_{δ_a} (1/s ²)	0.725	1.622	2.329
N_{δ_a} (1/s ²)	0.050	0.036	0.062

II. Short Period, Long Period, and Dutch Roll Approximation at Sea Level Standard

a. Short Period Approximation:

(i) Short-Period Approximation

$$\begin{bmatrix} \dot{\Delta \alpha} \\ \Delta q \end{bmatrix} = \begin{bmatrix} Z_{\alpha}/U^* & 1 \\ M_{\alpha} + M_{\dot{\alpha}} \frac{Z_{\alpha}}{U^*} & M_q + M_{\dot{q}} \end{bmatrix} \begin{bmatrix} \Delta \alpha \\ \Delta q \end{bmatrix} + \begin{bmatrix} Z_{\delta e}/U^* \\ M_{\delta e} + M_{\dot{\alpha}} \frac{Z_{\delta e}}{U^*} \end{bmatrix} \Delta \delta e$$

Laplace transformation:

$$(1) s \Delta \alpha(s) = \frac{Z_{\alpha}}{U^*} \Delta \alpha(s) + \Delta q(s) + \frac{Z_{\delta e}}{U^*} \Delta \delta e(s)$$

$$(2) s \Delta q(s) = (M_{\alpha} + \frac{M_{\dot{\alpha}} Z_{\alpha}}{U^*}) \Delta \alpha(s) + (M_q + M_{\dot{q}}) \Delta q(s) + (M_{\delta e} + \frac{M_{\dot{\alpha}} Z_{\delta e}}{U^*}) \Delta \delta e(s)$$

divide by $\Delta \delta e(s)$

$$\text{from (1): } (s - \frac{Z_{\alpha}}{U^*}) \frac{\Delta \alpha(s)}{\Delta \delta e(s)} - \frac{\Delta q(s)}{\Delta \delta e(s)} = \frac{Z_{\delta e}}{U^*} \frac{\Delta \delta e(s)}{\Delta \delta e(s)} \quad (3)$$

$$\text{from (2): } - (M_{\alpha} + \frac{M_{\dot{\alpha}} Z_{\alpha}}{U^*}) \frac{\Delta \alpha(s)}{\Delta \delta e(s)} + (s - (M_q + M_{\dot{q}})) \frac{\Delta q(s)}{\Delta \delta e(s)} = (M_{\delta e} + \frac{M_{\dot{\alpha}} Z_{\delta e}}{U^*}) \frac{\Delta \delta e(s)}{\Delta \delta e(s)} \quad (4)$$

write 3 and 4 as a matrix

$$\begin{bmatrix} s - \frac{Z_{\alpha}}{U^*} & -1 \\ -(M_{\alpha} + \frac{M_{\dot{\alpha}} Z_{\alpha}}{U^*}) & s - (M_q + M_{\dot{q}}) \end{bmatrix} \begin{bmatrix} \frac{\Delta \alpha(s)}{\Delta \delta e(s)} \\ \frac{\Delta q(s)}{\Delta \delta e(s)} \end{bmatrix} = \begin{bmatrix} \frac{Z_{\delta e}}{U^*} \\ M_{\delta e} + \frac{M_{\dot{\alpha}} Z_{\delta e}}{U^*} \end{bmatrix}$$

$$\begin{bmatrix} \frac{\Delta \alpha(s)}{\Delta \delta e(s)} \\ \frac{\Delta q(s)}{\Delta \delta e(s)} \end{bmatrix} = A^{-1} \begin{bmatrix} \frac{Z_{\delta e}}{U^*} \\ M_{\delta e} + \frac{M_{\dot{\alpha}} Z_{\delta e}}{U^*} \end{bmatrix} \quad \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} \frac{\Delta \alpha(s)}{\Delta \delta e(s)} \\ \frac{\Delta q(s)}{\Delta \delta e(s)} \end{bmatrix} = \frac{1}{\det(A)} \begin{bmatrix} s - (M_q + M_{\dot{q}}) & 1 \\ M_{\alpha} + \frac{M_{\dot{\alpha}} Z_{\alpha}}{U^*} & s - \frac{Z_{\alpha}}{U^*} \end{bmatrix} \begin{bmatrix} \frac{Z_{\delta e}}{U^*} \\ M_{\delta e} + \frac{M_{\dot{\alpha}} Z_{\delta e}}{U^*} \end{bmatrix}$$

$$= \frac{1}{\det(A)} \begin{bmatrix} \frac{Z_{\delta e}}{U^*} s - \frac{Z_{\delta e}}{U^*} (M_q + M_{\dot{q}}) + M_{\delta e} + \frac{M_{\dot{\alpha}} Z_{\delta e}}{U^*} \\ \frac{Z_{\delta e}}{U^*} (M_{\alpha} + \frac{M_{\dot{\alpha}} Z_{\alpha}}{U^*}) + (M_{\delta e} + \frac{M_{\dot{\alpha}} Z_{\delta e}}{U^*}) (s - \frac{Z_{\alpha}}{U^*}) \end{bmatrix}$$

$$\det(A) = (s - \frac{Z_{\alpha}}{U^*}) (s - (M_q + M_{\dot{q}})) - (M_{\alpha} + \frac{M_{\dot{\alpha}} Z_{\alpha}}{U^*})$$

$$= s^2 - (\frac{Z_{\alpha}}{U^*} + M_q + M_{\dot{q}}) s + \frac{Z_{\alpha}}{U^*} (M_q + M_{\dot{q}}) - M_{\alpha} - \frac{M_{\dot{\alpha}} Z_{\alpha}}{U^*}$$

Transfer Functions:

$$\frac{\Delta \alpha(s)}{\Delta \delta e(s)} = \frac{(\frac{Z_{\delta e}}{U^*}) s + M_{\delta e} - M_q \frac{Z_{\delta e}}{U^*}}{s^2 - (\frac{Z_{\alpha}}{U^*} + M_q + M_{\dot{q}}) s + (\frac{Z_{\alpha}}{U^*} M_q - M_{\alpha})}$$

$$\frac{\Delta q(s)}{\Delta \delta e(s)} = \frac{(M_{\delta e} + \frac{M_{\dot{\alpha}} Z_{\delta e}}{U^*}) s + (M_{\alpha} \frac{Z_{\delta e}}{U^*} - M_{\delta e} \frac{Z_{\alpha}}{U^*})}{s^2 - (\frac{Z_{\alpha}}{U^*} + M_q + M_{\dot{q}}) s + (\frac{Z_{\alpha}}{U^*} M_q - M_{\alpha})}$$

Plug in values from Appendix H:

$$\frac{\Delta \alpha(s)}{\Delta \delta e(s)} = \frac{-0.0419s - 1.384}{s^2 + 1.681s + 2.616} \quad (5)$$

$$\frac{\Delta q(s)}{\Delta \delta e(s)} = \frac{-1.34s - 0.761}{s^2 + 1.681s + 2.616} \quad (6)$$

b. Long Period Approximation:

Long-Period:

$$\begin{bmatrix} \Delta \dot{U} \\ \Delta \dot{\theta} \end{bmatrix} = \begin{bmatrix} x_u & -g \\ -\frac{z_u}{U^*} & 0 \end{bmatrix} \begin{bmatrix} \Delta U \\ \Delta \theta \end{bmatrix} + \begin{bmatrix} x_{\delta e} & x_{\delta r} \\ -\frac{z_{\delta e}}{U^*} & -\frac{z_{\delta r}}{U^*} \end{bmatrix} \begin{bmatrix} \Delta \delta e \\ \Delta \delta r \end{bmatrix}$$

Laplace:

$$(7) \quad s \Delta U(s) = x_u \Delta U(s) - g \Delta \theta(s) + x_{\delta e} \Delta \delta e(s) + x_{\delta r} \Delta \delta r(s)$$

$$(8) \quad s \Delta \theta(s) = -\frac{z_u}{U^*} \Delta U(s) - \frac{z_{\delta e}}{U^*} \Delta \delta e(s) - \frac{z_{\delta r}}{U^*} \Delta \delta r(s)$$

i) Let $\Delta \delta e = 0$

$$(7): (s - x_u) \frac{\Delta U(s)}{\Delta \delta r(s)} + g \frac{\Delta \theta(s)}{\Delta \delta r(s)} = x_{\delta r} \frac{\Delta \delta r(s)}{\Delta \delta r(s)} \quad (9)$$

$$(8): \frac{z_u}{U^*} \frac{\Delta U(s)}{\Delta \delta r(s)} + s \frac{\Delta \theta(s)}{\Delta \delta r(s)} = -\frac{z_{\delta r}}{U^*} \frac{\Delta \delta r(s)}{\Delta \delta r(s)} \quad (10)$$

$$\begin{bmatrix} \frac{\Delta U(s)}{\Delta \delta r(s)} \\ \frac{\Delta \theta(s)}{\Delta \delta r(s)} \end{bmatrix} = \underbrace{\begin{bmatrix} s - x_u & g \\ \frac{z_u}{U^*} & 0 \end{bmatrix}}_{A^{-1}} \begin{bmatrix} x_{\delta r} \\ -\frac{z_{\delta r}}{U^*} \end{bmatrix}$$

$$\det(A) = s^2 + x_u s - g \frac{z_u}{U^*}$$

$$\begin{bmatrix} \frac{\Delta U(s)}{\Delta \delta r(s)} \\ \frac{\Delta \theta(s)}{\Delta \delta r(s)} \end{bmatrix} = \frac{1}{\det(A)} \begin{bmatrix} s & -g \\ -\frac{z_u}{U^*} & s - x_u \end{bmatrix} \begin{bmatrix} x_{\delta r} \\ -\frac{z_{\delta r}}{U^*} \end{bmatrix}$$

$$= \frac{1}{\det(A)} \begin{bmatrix} x_{\delta r} s + g \frac{z_{\delta r}}{U^*} \\ -\frac{z_u}{U^*} x_{\delta r} - \left(\frac{z_{\delta r}}{U^*} \right) s + \frac{x_u}{U^*} \end{bmatrix}$$

$$\frac{\Delta U(s)}{\Delta \delta r(s)} = \frac{x_{\delta r} s + g \frac{z_{\delta r}}{U^*}}{s^2 + x_u s - g \frac{z_u}{U^*}} \quad (11)$$

$$\frac{\Delta \theta(s)}{\Delta \delta r(s)} = \frac{\left(-\frac{z_{\delta r}}{U^*} \right) s + \frac{z_{\delta r}}{U^*} \frac{x_u}{U^*} - \frac{z_u}{U^*} \frac{x_{\delta r}}{U^*}}{s^2 + x_u s - g \frac{z_u}{U^*}}$$

Plug in values from Appendix H:

$$x_{\delta r} = \frac{P_{max}}{m U^*}$$

Pratt & Whitney JT3D-7, 4 engines

$$T_{total} = 4 \times 19000 \text{ lbf} = 76000 \text{ lbf}$$

$$P_{max} = T_{total} U^* = 76000 \times 243 = 1.85 \times 10^7 \text{ lbf} \cdot \text{ft} \cdot \text{s}^{-1}$$

$$x_{\delta r} = \frac{1.85 \times 10^7 \left[\frac{\text{ft} \cdot \text{lbf}}{\text{s}} \right]}{\frac{190000 \text{ [slug]} \times 243 \left[\frac{\text{ft}}{\text{s}} \right]}} = 12.9 \left[\frac{\text{lbf}}{\text{slug}} \right] \quad [\text{lbf}] = \left[\frac{\text{slug} \cdot \text{ft}}{\text{s}^2} \right]$$

$$x_{\delta r} = 12.9 \left[\frac{\text{ft}}{\text{s}^2} \right]$$

$$\frac{\Delta U(s)}{\Delta \delta e(s)} = \frac{12.9 s}{s^2 + 0.0291s + 0.03326}$$

$$\frac{\Delta \theta(s)}{\Delta \delta e(s)} = 0$$

ii) $\Delta \delta r = 0$

$$\frac{\Delta U(s)}{\Delta \delta e(s)} = \frac{x_{\delta e} s + g \frac{z_{\delta e}}{U^*}}{s^2 + x_u s - g \frac{z_u}{U^*}} = \frac{-0.0419 g}{s^2 + 0.0291s + 0.03326}$$

$$\frac{\Delta \theta(s)}{\Delta \delta e(s)} = \frac{\left(-\frac{z_{\delta e}}{U^*} \right) s + \frac{z_{\delta e}}{U^*} \frac{x_u}{U^*} - \frac{z_u}{U^*} \frac{x_{\delta e}}{U^*}}{s^2 + x_u s - g \frac{z_u}{U^*}}$$

$$\frac{\Delta \theta(s)}{\Delta \delta e(s)} = \frac{-0.0419 s + 1.218 \times 10^{-3}}{s^2 + 0.0291s + 0.03326}$$

c. Dutch Roll Approximation:

Dutch Roll Approximation		McDonnell Douglas DC-8	
$\Delta p(s) = \frac{Y_p}{U^*} \Delta p(s) - \left(1 - \frac{Y_c}{U^*}\right) \Delta r(s) + \frac{Y_{\delta_r}}{U^*} \Delta \delta_r(s)$		Aircraft DC-8	
$\Delta r(s) = N_p \Delta p(s) + N_r \Delta r(s) + N_{\delta_r} \Delta \delta_r(s) + N_{\delta_a} \Delta \delta_a(s)$		Parameter Assume $\alpha_{trim} = -2^\circ$	
1) $s \Delta p(s) = \frac{Y_p}{U^*} \Delta p(s) - \left(1 - \frac{Y_c}{U^*}\right) \Delta r(s) + \frac{Y_{\delta_r}}{U^*} \Delta \delta_r(s)$		Altitude (ft) S/L	
2) $s \Delta r(s) = N_p \Delta p(s) + N_r \Delta r(s) + N_{\delta_r} \Delta \delta_r(s) + N_{\delta_a} \Delta \delta_a(s)$		Mach 0.218	
3) $\left(s - \frac{Y_p}{U^*}\right) \Delta p(s) - \left(\frac{Y_c}{U^*} - 1\right) \Delta r(s) = \frac{Y_{\delta_r}}{U^*} \Delta \delta_r(s)$		True Airspeed (ft/s) 243	
4) $(s - N_r) \Delta r(s) - N_p \Delta p(s) = N_{\delta_r} \Delta \delta_r(s) + N_{\delta_a} \Delta \delta_a(s)$		Dyn Pressure (lb/ft ²) 70.4	
5) $\left(s - \frac{Y_p}{U^*}\right) \frac{\Delta p(s)}{\Delta \delta_r(s)} - \left(\frac{Y_c}{U^*} - 1\right) \frac{\Delta r(s)}{\Delta \delta_r(s)} = \frac{Y_{\delta_r}}{U^*}$		Weight (lb) 190,000	
6) $(s - N_r) \frac{\Delta r(s)}{\Delta \delta_r(s)} - N_p \frac{\Delta p(s)}{\Delta \delta_r(s)} = N_{\delta_r}$		Wing Area—S—(ft ²) 2,600.0	
$\bar{A} \begin{bmatrix} s - \frac{Y_p}{U^*} & 1 - \frac{Y_c}{U^*} \\ -N_p & s - N_r \end{bmatrix} \begin{bmatrix} \frac{\Delta p(s)}{\Delta \delta_r(s)} \\ \frac{\Delta r(s)}{\Delta \delta_r(s)} \end{bmatrix} = \begin{bmatrix} \frac{Y_{\delta_r}}{U^*} \\ N_{\delta_r} \end{bmatrix}$		Wing Span—b—(ft) 142.3	
		Wing Chord—c—(ft) 23.0	
		C.G. (x c) 0.15	
		Trim AOA (deg) 10.36	
		I_{xx} (slug-ft ²) 3.16×10^6	
		I_{yy} (slug-ft ²) 2.94×10^6	
		I_{zz} (slug-ft ²) 5.51×10^6	
		I_{xzy} (slug-ft ²) -4.14×10^5	
		Longitudinal Derivatives	
		X_u (1/s) -0.0291	
		X_a (ft/s ²) 15.32	
		Z_u (1/s) -0.251	
		Z_a (ft/s ²) -152.8	
		M_u (1/ft-s) 0.0000	
		M_a (1/s ²) -2.118	
		M_z (1/s) -0.260	
		$M_{\dot{z}}$ (1/s) -0.792	
		$X_{\dot{z}}$ (ft/s ²) 0.0	
		$Z_{\dot{z}}$ (ft/s ²) -10.17	
		$M_{\dot{z}}$ (1/s ²) -1.351	
		Lat-Dir Derivatives	
		Y_p (ft/s ²) -27.05	
		L_p (1/s ²) -1.334	
		L_r (1/s) -0.949	
		$L_{\dot{r}}$ (1/s) 0.611	
		N_p (1/s ²) 0.762	
		N_r (1/s) -0.119	
		$N_{\dot{r}}$ (1/s) -0.268	
		616 INTRODUCTION TO AIF	
		Y_{δ_r} (ft/s ²) 5.782	
		L_{δ_r} (1/s ²) 0.185	
		N_{δ_r} (1/s ²) -0.389	
		Y_{δ_a} (ft/s ²) 0.0	
		L_{δ_a} (1/s ²) 0.725	
		N_{δ_a} (1/s ²) 0.050	
$\det \bar{A} = s^2 - \left(\frac{Y_p}{U^*} + N_r\right)s + \frac{Y_p N_r}{U^*} + N_p - \frac{N_p Y_c}{U^*}$			
$s^2 - \left(\frac{-27.05}{243} - 0.268\right)s + \frac{27.05 \cdot -0.268}{243} + 0.762 - \frac{-0.268 \cdot 0}{243}$			
Char eq: $s^2 + 0.3793s + 0.7918$			
$\frac{\Delta p(s)}{\Delta \delta_r(s)} = \frac{\frac{Y_{\delta_r}}{U^*} s - \frac{N_r Y_{\delta_r}}{U^*} + \frac{Y_c N_{\delta_r}}{U^*} - N_{\delta_r}}{s^2 - \left(\frac{Y_p}{U^*} + N_r\right)s + \frac{Y_p N_r}{U^*} + N_p - \frac{N_p Y_c}{U^*}}$			
$\frac{\Delta p(s)}{\Delta \delta_r(s)} = \frac{0.02379s - 0.3953}{s^2 + 0.3793s + 0.7918}$			
$\frac{\Delta r(s)}{\Delta \delta_r(s)} = \frac{N_{\delta_r} s + \frac{N_p Y_{\delta_r}}{U^*} - \frac{Y_p N_{\delta_r}}{U^*}}{s^2 - \left(\frac{Y_p}{U^*} + N_r\right)s + \frac{Y_p N_r}{U^*} + N_p - \frac{N_p Y_c}{U^*}}$			
$\frac{\Delta r(s)}{\Delta \delta_r(s)} = \frac{-0.369s - 0.02517}{s^2 + 0.3793s + 0.7918}$			

III. Poles, Natural Frequency, and Damping Ratio

a) Short Period:

Short Period:

$$0 = \lambda^2 + \lambda \left(\underbrace{-M\dot{q} - M\ddot{x}}_b - \frac{Z\alpha}{U^*} \right) + \underbrace{\frac{Z\alpha}{U^*} M\dot{q} - M\alpha}_c$$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$a = 1 \quad b = 1.6808$$

$$c = 2.616$$

$$\lambda_{1,2} = \frac{-1.6808 \pm \sqrt{1.6808^2 - 4 \times 2.616}}{2} = -0.8404 \pm j 1.3819$$

$$\bar{\omega}_{nsp} = \sqrt{0.8404^2 + 1.3819^2} = 1.6174 \text{ rad s}^{-1}$$

$$\zeta_{sp} = \frac{-\text{Re}}{\omega_{nsp}} = \frac{0.8404}{1.6174} = 0.5196$$

b) Long Period:

Long Period:

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-0.0291 \pm \sqrt{0.0291^2 - 4 \times 1 \times \frac{32.2 \times 0.2}{243}}}{2}$$

$$= \frac{-0.0291 \pm j 0.363584267}{2}$$

$$\lambda_{1,2} = -0.01455 \pm j 0.1817921335$$

$$\bar{\omega}_{nlp} = \sqrt{(-0.01455)^2 + (0.1817921335)^2}$$

$$\bar{\omega}_{nlp} = 0.1823734693 \text{ rad s}^{-1}$$

$$\zeta_{lp} = \frac{-(-0.01455)}{\bar{\omega}_{nlp}} = 0.0797813413$$

c) Dutch Roll:

Dutch Roll

$$\lambda^2 + \underbrace{(-N_r - \frac{Y_\beta}{U^*})}_b \lambda + \underbrace{N_\beta + \frac{Y_\beta}{U^*} N_r}_c$$

$$b = 0.268 + \frac{27.85}{243} = 0.3893$$

$$c = 0.7918$$

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-0.3893 \pm \sqrt{0.3893^2 - 4 \times 0.7918}}{2}$$

$$\lambda_{1,2} = -0.1897 \pm j 0.8694$$

$$\text{Den}_{DR}(s) = s^2 + 0.3893s + 0.7918$$

$$\text{TF}_{DR}(s) = \frac{K_{DR}}{s^2 + 0.3893s + 0.7918}$$

$$\omega_{nDR} = \sqrt{(-0.1897)^2 + 0.8694^2} = 0.8899$$

$$\zeta_{DR} = -\frac{\text{Re}}{\omega_{nDR}} = \frac{0.1897}{0.8899} = 0.2132$$

IV. Angle of Attack PID Controller Design

a. Characteristic Polynomial

$$C(s) = \frac{Y(s)}{R(s)} = \frac{G_c(s) G_A(s) G(s)}{1 + G(s) H(s)}$$

$$G_c(s) = K_d s + K_p + \frac{K_I}{s} = \frac{K_d s^2 + K_p s + K_I}{s}$$

$$C(s) = \frac{K_d s^2 + K_p s + K_I}{s} \times \frac{50}{s+50} \times \frac{-0.0419s - 1.384}{s^2 + 1.681s + 2.616}$$

$$1 + \frac{K_d s^2 + K_p s + K_I}{s} \times \frac{50}{s+50} \times \frac{-0.0419s - 1.384}{s^2 + 1.681s + 2.616}$$

$$C(s) = \frac{(K_d s^2 + K_p s + K_I) \times (-2.095s - 69.2)}{s(s+50)(s^2 + 1.681s + 2.616) + (K_d s^2 + K_p s + K_I)(-2.095s - 69.2)}$$

$$s(s+50)(s^2 + 1.681s + 2.616) + (K_d s^2 + K_p s + K_I)(-2.095s - 69.2)$$

$$\text{num} = -2.095 K_d s^3 - 69.2 K_d s^2 - 2.095 K_p s^2 - 69.2 K_p s - 2.095 K_I s - 69.2 K_I$$

$$\text{num} = -2.095 K_d s^3 + (-69.2 K_d - 2.095 K_p) s^2 + (-69.2 K_p - 2.095 K_I) s - 69.2 K_I$$

$$\text{den} = s(s^2 + 1.681s + 2.616) + 50s^2 + 84.05s + 130.8$$

$$-2.095 K_d s^3 + (-69.2 K_d - 2.095 K_p) s^2 + (-69.2 K_p - 2.095 K_I) s - 69.2 K_I$$

$$\text{den} = s^4 + 1.681 s^3 + 26.666 s^2 + 130.8 s$$

$$-2.095 K_d s^3 + (-69.2 K_d - 2.095 K_p) s^2 + (-69.2 K_p - 2.095 K_I) s - 69.2 K_I$$

$M_n (1/(n-1))$	$M_n (1/n^2)$
$M_1 (1/s)$	-2.118
$M_2 (1/s^2)$	-0.260
$M_3 (1/s^3)$	-0.792
$M_4 (1/s^4)$	0.0
$M_5 (1/s^5)$	-10.17
$M_6 (1/s^6)$	-1.351

$$\text{den} = s^4 + (51.681 - 2.095k_d)s^3 + (86.666 - 69.2k_d - 2.095k_p)s^2 + (150.8 - 69.2k_p - 2.095k_r)s - 69.2k_r$$

Desired:

$$M_p = 0.05$$

$$t_s < 10 \text{ s}$$

$$M_p = e^{-\frac{\pi \zeta}{\sqrt{1-\zeta^2}}}$$

$$\zeta_d = \frac{(\ln M_p)^2}{\pi^2 + (\ln M_p)^2} = \frac{(\ln 0.05)^2}{\pi^2 + (\ln 0.05)^2}$$

$$\zeta_d = 0.6901$$

$$t_s = \frac{3}{\zeta_d \omega_n} \Rightarrow \omega_n = \frac{3}{t_s \zeta_d} = \frac{3}{10 \times 0.6901}$$

$$\omega_n = 0.4347 \text{ rad s}^{-1}$$

$$p_3 = -10 \bar{\omega}_n \bar{\zeta}_d = -10 \times 0.4347 \times 0.6901 = -2.99486$$

$$p_3 = -3$$

Desired char. eq.

$$0 = (s^2 + 2\zeta_d \omega_n s + \omega_n^2)(s - p_3)(s + 50)$$

$$0 = (s^2 + 2 \times 0.6901 \times 0.4347 s + 0.4347^2)(s + 3)(s + 50)$$

$$0 = (s^2 + 0.6 s + 0.189)(s + 3)(s + 50)$$

$$0 = (s^3 + 3.6 s^2 + 1.989 s + 0.567)(s + 50)$$

$$0 = s^4 + 3.6 s^3 + 1.989 s^2 + 0.567 s + 50 s^3 + 180 s^2 + 99.45 s + 28.35$$

$$0 = s^4 + 53.6 s^3 + 181.989 s^2 + 100.017 s + 28.35$$

$$\begin{aligned}
 C(s) &= \frac{k_d s^2 + k_p s + k_I}{s} \times \frac{50}{s+50} \times \frac{-0.0419s - 1.384}{s^2 + 1.681s + 2.616} \\
 &= 1 + \frac{k_d s^2 + k_p s + k_I}{s} \times \frac{50}{s+50} \times \frac{-0.0419s - 1.384}{s^2 + 1.681s + 2.616} \\
 &= \frac{(k_d s^2 + k_p s + k_I)(-0.0419s - 1.384)}{s^3 + 1.681s^2 + 2.616s + (k_d s^2 + k_p s + k_I)(-0.0419s - 1.384)} \\
 \text{num} &= -0.0419k_d s^3 - 0.0419k_p s^2 - 0.0419k_I s - 1.384k_d s^2 - 1.384k_p s - 1.384k_I \\
 \text{num} &= -0.0419k_d s^3 + (-0.0419k_p - 1.384k_d)s^2 + (-0.0419k_I - 1.384k_p)s - 1.384k_I \\
 \text{den} &= (1 - 0.0419k_d)s^3 + (1.681 - 0.0419k_p - 1.384k_d)s^2 + (2.616 - 0.0419k_I - 1.384k_p)s - 1.384k_I \\
 D &= s^3 + \left(\frac{1.681 - 0.0419k_p - 1.384k_d}{1 - 0.0419k_d} \right) s^2 + \left(\frac{2.616 - 0.0419k_I - 1.384k_p}{1 - 0.0419k_d} \right) s + \left(\frac{-1.384k_I}{1 - 0.0419k_d} \right)
 \end{aligned}$$

Desired char. eq.

$$0 = (s^2 + 2\zeta_d \omega_{nd} s + \omega_{nd}^2)(s - p_3)$$

$$0 = (s^2 + 2 \times 0.6901 \times 0.4347 s + 0.4347^2)(s + 3)$$

$$0 = (s^2 + 0.6s + 0.189)(s + 3)$$

$$0 = s^3 + 3.6s^2 + 1.989s + 0.567$$

$$\frac{1.681 - 0.0419k_p - 1.384k_d}{1 - 0.0419k_d} = 3.6$$

$$\frac{2.616 - 0.0419k_I - 1.384k_p}{1 - 0.0419k_d} = 1.989$$

$$\frac{-1.384k_I}{1 - 0.0419k_d} = 0.567$$

$$\text{I)} \frac{1.681 - 0.0419 K_p - 1.384 K_d}{1 - 0.0419 K_d} = 3.6$$

$$\text{II)} \frac{2.616 - 0.0419 K_I - 1.384 K_p}{1 - 0.0419 K_d} = 1.989$$

$$\text{III)} \frac{-1.384 K_I}{1 - 0.0419 K_d} = 0.567$$

$$\text{Rearrange: I)} K_p = \frac{3.6(1 - 0.0419 K_d) + 1.384 K_d - 1.681}{-0.0419}$$

$$\text{III)} K_I = \frac{0.567(1 - 0.0419 K_d)}{-1.384}$$

I) & III) into II):

$$\text{II)} 2.616 - 0.0419 \left(\frac{0.567(1 - 0.0419 K_d)}{-1.384} \right) + 1.384 \left(\frac{3.6(1 - 0.0419 K_d) + 1.384 K_d - 1.681}{-0.0419} \right) = 1.989$$

$$K_d = -1.5688 = \boxed{-1.569}$$

K_d into III):

$$\text{III)} = -1.384 K_I = 0.567(1 - 0.0419(-1.569))$$

$$K_I = \frac{0.604275}{-1.384} = \boxed{-0.4366}$$

K_d into I):

$$\text{I)} K_p = \frac{3.6(1 - 0.0419(-1.569)) + 1.384(-1.569) - 1.681}{-0.0419}$$

$$K_p = \frac{3.2366 + (-2.17149) - 1.681}{-0.0419} = \boxed{0.372}$$

$$K_I = -0.4366$$

$$K_p = 0.372$$

$$K_d = -1.569$$

Characteristic Equation: $s^3 + 3.6s^2 + 1.989s + 0.567 = 0$

Table 1: Gain Value Verified with MATLAB

K_p	0.3718
K_I	-0.4366
K_d	-1.569

b. Final Value Theorem

iv)b) Use the Final Value Theorem to predict the steady-state value of $\Delta\alpha$ in response to a commanded step input of $\Delta\alpha_{com}(t) = 5$ deg and determine the steady-state error.

$$\Delta\alpha_{ss} = \lim_{s \rightarrow 0} s \Delta\alpha(s) \quad \Delta\alpha_{com} = \frac{5}{s}$$

$$\Delta\alpha(s) = C(s) \Delta\alpha_{com}(s) = \lim_{s \rightarrow 0} \left(\frac{-0.0419k_d s^3 + (-0.0419k_p - 1.384k_d)s^2 + (-0.0419k_f - 1.384k_p)s - 1.384k_f}{(1 - 0.0419k_d)s^3 + (1.681 - 0.0419k_p - 1.384k_d)s^2 + (2.616 - 0.0419k_f - 1.384k_p)s - 1.384k_f} \right) \frac{5}{s}$$

$$\Delta\alpha_{ss} = \left(\frac{-1.384k_f}{-1.384k_f} \right) 5 = 5$$

$$e_{ss} = \Delta\alpha_{com} - \Delta\alpha_{ss} = 5 - 5 = 0$$

$$\Delta\alpha_{ss} = 5$$

c. MATLAB Visualization

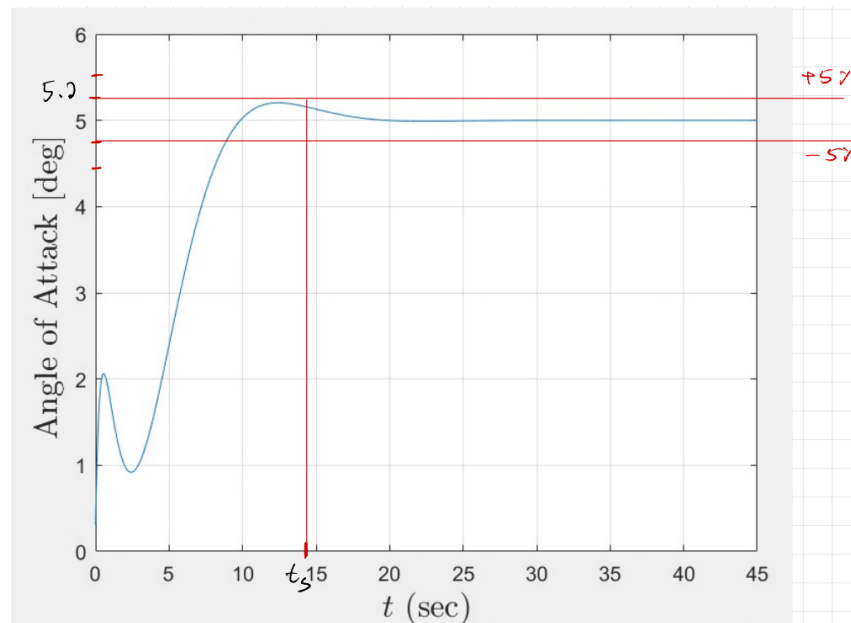


Figure 1: AOA Response with Designed PID Controller

The maximum overshoot percentage was calculated to be 4%, this is within the specified 5% overshoot limit, so the overshoot requirement is satisfied. The settling time of 15 seconds doesn't satisfy the settling time constraint of less than 10 seconds. The response converges exactly to the step values of 5 degrees, indicating that the PID controller tracks the step input without steady-state error.

d. Concluding Remarks

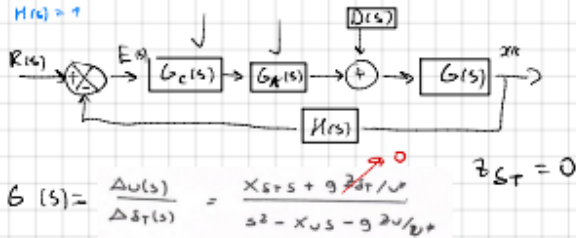
When the actuator is included in the system, the desired characteristic equation is to the 4th degree and has 4 poles. The unknown values to be solved are the k gain values. Because there are 3 unknowns and 4 equations, the system of equations is undetermined without a unique solution and will have infinitely many solutions. To avoid that, the actuator was assumed to be perfect, and a new closed loop transfer function was calculated. Since $G_A(s) = 1$, the desired characteristic equation now has 3 poles. As a result, the k gain values that were calculated from the simplified system are very small.

V. Throttle Input PID Controller Design

a. Characteristic Polynomial

v) Design a PID controller to control the forward speed of the selected aircraft using throttle input, assuming the sensor system is perfect and the throttle actuator transfer function is $G_A(s) = \frac{75}{s+75}$.

v)a) Compute a desired characteristic polynomial for the closed-loop system so that the complex conjugate pair of poles has a damping ratio and natural frequency corresponding to a maximum overshoot of 7% and a settling time of less than 15 sec. Choose the third, real-valued pole, as $p_3 = -10\omega_n\zeta$. Then compute the PID controller gains required to obtain the desired closed-loop system eigenvalues.



$$\frac{x_{ST} s}{s^2 + 0.0291s + 0.03326}$$

$$x_{ST} = \frac{p_{max}}{m U^*}$$

Pratt & Whitney JT3D-7, 4 engines

$$T_{total} = 4 \times 19000 \text{ lbf} = 76000 \text{ lbf}$$

$$p_{max} = T_{total} U^* = 76000 \times 243 = 1.85 \times 10^7 \text{ lbf} \cdot \text{ft} \cdot \text{s}^{-1}$$

$$x_{ST} = \frac{1.85 \times 10^7 \left[\frac{\text{ft} \cdot \text{lbf}}{\text{s}} \right]}{\frac{190000}{32.2} [\text{slug}] \times 243 \left[\frac{\text{ft}}{\text{s}} \right]} = 12.9 \left[\frac{\text{lbf}}{\text{slug}} \right] \quad (\text{lbf}) = \left[\frac{\text{slug} \cdot \text{ft}}{\text{s}^2} \right]$$

$$x_{ST} = 12.9 \left[\frac{\text{ft}}{\text{s}^2} \right] \Rightarrow G(s) = \frac{12.9 s}{s^2 + 0.0291s + 0.03326}$$

$$C(s) = \frac{G_c(s) G_A(s) G(s)}{1 + G_c(s) G_A(s) G(s) H(s)}$$

$$C(s) = \frac{\frac{K_d s^2 + K_p s + K_f}{s} \times \frac{75}{s+75} \times \frac{12.9 s}{s^2 + 0.0291s + 0.03326}}{1 + \frac{K_d s^2 + K_p s + K_f}{s} \times \frac{75}{s+75} \times \frac{12.9 s}{s^2 + 0.0291s + 0.03326}}$$

$$C(s) =$$

$$\frac{\text{num}}{\text{den}} = \frac{12.9 (K_d s^2 + K_p s + K_f) \times 75}{(s^3 + 0.0291s^2 + 0.03326s) (s+75) + 12.9 (K_d s^3 + K_p s^2 + K_f s) \times 75}$$

Aircraft Parameter	DC-8
Assume $\alpha_{L-8} = -2^\circ$	
Altitude (ft)	S/L
Mach	0.218
True Airspeed (ft/s)	243
Dyn Pressure (lb/ft ²)	70.4
Weight (lb)	190,000
Wing Area—S—(ft ²)	2,690.0
Wing Span—b—(ft)	142.3
Wing Chord—c—(ft)	23.0
C.G. (x c)	0.15
Trim AOA (deg)	10.36
I_{xx} (slug-ft ²)	3.16×10^6
I_{yy} (slug-ft ²)	2.94×10^6
I_{zz} (slug-ft ²)	5.51×10^6
I_{xy} (slug-ft ²)	-4.14×10^5
Longitudinal Derivatives	
X_1 (1/s)	-0.0291
X_2 (ft/s ²)	15.32
X_3 (1/s)	-0.251
Z_1 (ft/s ²)	-152.8
Z_2 (ft/s)	0.0000
M_1 (1/(ft-s))	-2.118
M_2 (1/s ²)	-0.260
M_3 (1/s)	-0.792
A_1 (ft/s ²)	0.0
A_2 (ft/s ²)	-10.17
A_3 (1/s ²)	-1.351

$$\text{den} = s^4 + (968.5 k_d + 75.0291) s^3 + (968.5 k_p + 2.21576) s^2 + (968.5 k_z + 2.495) s$$

Desired char. eq.

$$0 = (s^2 + 2 \zeta_d \omega_{nd} s + \omega_{nd}^2) (s - p_3) (s + 75)$$

$$M_p = 0.07$$

$$t_s < 15 s$$

$$M_p = e^{-\frac{\pi \zeta_d}{\sqrt{1-\zeta_d^2}}}$$

$$p_3 = -10 \zeta_d \omega_{nd}$$

$$p_3 = -1.99967$$

$$p_3 \approx -2$$

$$\zeta_d = \frac{(\ln M_p)^2}{\pi^2 + (\ln M_p)^2} = \frac{(\ln 0.07)^2}{\pi^2 + (\ln 0.07)^2}$$

$$\zeta_d = 0.6461$$

$$t_s = \frac{3}{\zeta_d \omega_{nd}} \Rightarrow \omega_{nd} = \frac{3}{\zeta_d t_s} = \frac{3}{0.6461 \times 15}$$

$$\omega_{nd} = 0.3095 \text{ rad/s}$$

$$0 = (s^2 + 0.45 + 0.0958) (s + 2) (s + 75)$$

$$0 = (s^3 + 2.45s^2 + 0.8958s + 0.1916) (s + 75)$$

$$0 = s^4 + 99.5s^3 + 1800.9s^2 + 67.38s + 14.37$$

The closed-loop characteristic equation is missing the constant term when the desired characteristic equation has a 4th pole of $s = -75$ from the actuator. This makes it impossible to equate the terms and solving for the k gain values.

Instead, the 4th pole was removed from the desired char. eq. to make it possible to find k values.

$$C(s) = \frac{k_d s^2 + k_p s + k_z}{1 + \frac{k_d s^2 + k_p s + k_z}{s + 75} \times \frac{12.9}{s^2 + 0.0291s + 0.03326}}$$

$$C(s) =$$

$$\frac{\text{num}}{\text{den}} = \frac{12.9(k_d s^2 + k_p s + k_z) \times 75}{(s^2 + 0.0291s + 0.03326)(s + 75) + 12.9(k_d s^2 + k_p s + k_z) \times 75}$$

$$0 = (s^2 + 0.45 + 0.0958) (s + 2)$$

$$0 = (s^3 + 2.45s^2 + 0.8958s + 0.1916)$$

$$968.5 k_d + 75.0291 = 2.4$$

$$968.5 k_p + 2.21576 = 0.8958$$

$$968.5 k_z + 2.495 = 0.1916$$

$$k_d = -0.075$$

$$k_p = -7.364 \times 10^{-3}$$

$$k_z = -2.381 \times 10^{-3}$$

b. Final Value Theorem

$$\begin{aligned}
 b) \Delta u_{ss} &= \lim_{s \rightarrow 0} s \Delta u(s) & \Delta u_{com} &= \frac{10}{s} \\
 \Delta u(s) &= W(s) \Delta u_{com}(s) \\
 W(s) &= \frac{G(s)}{1 + G(s)G_d(s)G(s)H(s)} \\
 W(s) &= \frac{12.9s}{s^2 + 0.0291s + 0.03326} \\
 &= \frac{12.9s}{1 + \frac{k_d s^2 + k_p s + k_y}{s} \times \frac{75}{s+75} \times \frac{12.9s}{s^2 + 0.0291s + 0.03326}} \\
 &= \frac{12.9s}{s^2 + 0.0291s + 0.03326 + \frac{(k_d s^2 + k_p s + k_y) \times 967.5}{s+75}} \\
 W(s) &= \frac{12.9s(s+75)}{(s^2 + 0.0291s + 0.03326)(s+75) + (k_d s^2 + k_p s + k_y) \times 967.5} \\
 \Delta u_{ss} &= \lim_{s \rightarrow 0} s \Delta u(s) = \lim_{s \rightarrow 0} s \frac{12.9s(s+75)}{(s^2 + 0.0291s + 0.03326)(s+75) + (k_d s^2 + k_p s + k_y) \times 967.5} = 0 \\
 \Delta u_{ss} &= 0
 \end{aligned}$$

c. MATLAB Visualization

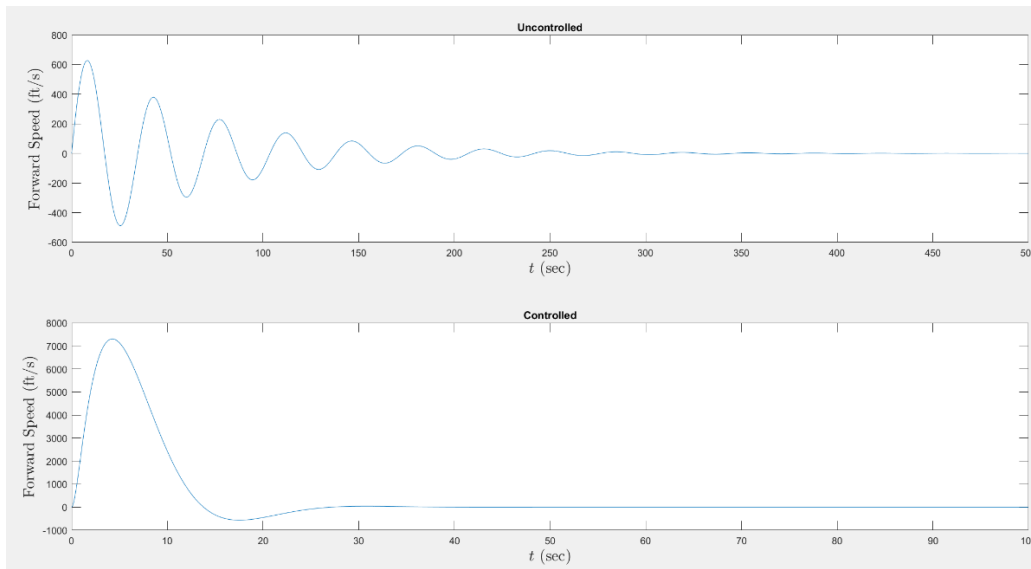


Figure 2: Forward Speed Response for uncontrolled system and controlled system with Designed PID Controller

d. Concluding Remarks

When calculating the closed loop characteristic equation, it was missing the constant term while the desired characteristic equation had a 4th pole of -75 from the actuator. This made it impossible to equate the terms and solving for the k gain values. Instead, the 4th pole was removed from the desired characteristic equation to make it possible to find the k values. As a result, the k values came out to be very small, meaning the actuator may not have been even needed. From figure 2, the controlled graph has a very high value of forward speed meaning that the initial conditions of a max. overshoot of 7% and a setting time of less than 15 seconds are not acceptable. The shapes for the controlled and uncontrolled graphs are similar in shape but very small k values and large forward speed value make the aircraft not feasible.